

AUCKLAND, New Zealand

WMO: 931100

Lat: 37.0089S

Lon: 174.8072E

Elev: 23

StdP: 14.68

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 39.4	(c) 41.6	(d) 35.4	(e) 30.3	(f) 43.2	(g) 37.3	(h) 32.7	(i) 44.8	(j) 31.3	(k) 54.2	(l) 28.0	(m) 54.9	(n) 3.5	(o) 160	(p) 0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.6	78.5	68.1	76.7	67.2	75.1	66.2	70.7	74.9	69.3	73.5	68.2	72.5	12.6	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.5	108.6	72.9	67.9	102.7	71.7	66.3	97.1	70.7	34.5	74.5	33.4	73.8	32.4	72.8	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.7	24.6	22.0		DB	35.7	81.9	1.2	1.9	34.8	83.3	34.2	84.4	33.5	85.5	32.7	86.8
				WB	34.7	73.0	1.1	1.6	33.9	74.1	33.3	75.1	32.7	76.0	31.9	77.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.0	67.8	68.7	66.2	62.0	57.8	53.7	52.1	53.0	55.6	57.9	60.9	65.1	(6)
	DBStd	6.59	3.27	2.94	3.10	3.57	3.99	4.17	3.69	3.12	3.16	2.94	3.13	3.27	(7)
	HDD50	51	0	0	0	0	2	15	22	10	2	0	0	0	(8)
	HDD65	2149	12	4	24	102	225	340	399	372	283	219	130	39	(9)
	CDD50	3706	551	525	502	361	243	125	88	103	169	246	326	467	(10)
	CDD65	327	98	109	61	12	1	0	0	0	0	0	5	41	(11)
	CDH74	537	202	224	61	3	0	0	0	0	0	0	2	45	(12)
	CDH80	13	5	7	1	0	0	0	0	0	0	0	0	0	(13)
(14) Wind	WSAvg	10.5	10.8	9.9	9.5	8.9	9.4	9.5	9.5	10.0	11.2	12.8	12.3	11.6	(14)
(15) Precipitation	PrecAvg	44.9	2.8	3.0	3.2	3.7	4.0	4.8	4.9	4.6	3.9	3.2	3.2	3.5	(15)
	PrecMax	58.4	7.9	9.1	7.6	8.8	7.5	10.8	7.8	8.6	9.0	7.9	9.2	8.0	(16)
	PrecMin	32.2	0.2	0.2	0.5	0.5	1.7	2.4	1.4	1.4	1.9	0.7	0.4	0.8	(17)
	PrecStd	7.4	2.1	2.5	1.9	1.7	1.6	1.9	1.6	1.9	1.3	1.7	1.7	1.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.6	80.9	77.4	73.8	69.7	64.8	62.9	63.0	66.4	68.7	73.5	77.3	(19)
		MCWB	68.8	70.2	66.8	66.3	64.3	61.4	58.7	57.6	59.4	60.9	64.9	67.2	(20)
	2%	DB	78.4	78.7	75.5	71.7	67.6	62.8	60.9	61.1	64.4	66.5	70.3	75.2	(21)
		MCWB	68.1	68.6	66.0	64.9	62.8	59.4	57.0	56.4	58.7	59.6	62.9	65.9	(22)
	5%	DB	76.6	76.9	73.7	69.9	65.9	61.1	59.3	59.9	62.7	64.8	68.3	73.3	(23)
		MCWB	67.0	67.7	65.1	63.8	61.7	57.9	55.6	55.4	57.6	58.7	61.4	64.9	(24)
	10%	DB	74.7	75.1	72.0	68.2	64.3	60.4	58.2	58.9	61.1	63.2	66.5	71.4	(25)
		MCWB	66.0	66.5	64.6	62.7	60.5	57.1	54.7	54.6	56.6	58.2	60.3	64.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	72.8	70.4	69.2	65.7	63.0	60.1	59.7	61.6	62.7	66.3	70.0	(27)
		MCDB	76.7	77.0	73.4	71.7	67.9	64.6	61.8	61.7	64.6	66.2	70.8	74.1	(28)
	2%	WB	70.2	71.2	68.8	67.1	63.9	60.3	58.0	57.8	59.9	61.3	64.6	68.2	(29)
		MCDB	75.1	75.3	72.5	69.9	66.2	62.1	60.2	60.2	62.9	64.7	68.6	72.0	(30)
	5%	WB	68.9	69.8	67.3	65.4	62.5	59.0	56.6	56.4	58.8	60.1	63.2	66.9	(31)
		MCDB	73.6	74.1	71.6	68.5	65.0	61.0	59.0	59.1	61.7	63.6	67.0	71.0	(32)
	10%	WB	67.4	68.6	65.9	63.8	60.8	57.3	55.2	55.2	57.4	59.2	61.7	65.4	(33)
		MCDB	72.2	73.0	70.7	67.2	63.7	59.7	57.8	58.1	60.5	62.7	65.8	69.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	12.6	12.7	12.2	11.3	11.0	11.2	11.2	11.1	10.7	11.3	11.4	(35)
		MCDBR	14.8	14.5	14.1	12.9	11.6	10.5	11.1	11.9	12.1	12.4	13.3	13.9	(36)
	5% WB	MCWBR	7.1	6.9	7.1	7.6	7.9	8.1	8.2	8.2	8.1	8.0	8.1	7.3	(37)
		MCWBR	12.1	11.7	11.7	10.3	9.7	9.5	9.8	10.4	10.6	11.0	11.7	11.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.351	0.351	0.329	0.318	0.312	0.308	0.303	0.313	0.334	0.342	0.338	0.349	(40)
		taud	2.546	2.568	2.633	2.653	2.639	2.629	2.640	2.606	2.524	2.507	2.544	2.555	(41)
	Ebn at Noon		312	305	299	284	267	259	268	282	292	304	314	314	(42)
		Edh at Noon	34	32	28	24	21	20	21	25	30	34	34	34	(43)
(44) All-Sky Solar Radiation	RadAvg		2126	1839	1537	1091	753	607	676	931	1265	1634	1930	2018	(44)
	RadStd		109	134	85	67	47	43	43	53	73	102	104	157	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-139	+178	N/A

Nomenclature: See separate page